

Asad Baig

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SUMMARY

Backend / AI Engineer experienced in async Python and FastAPI services, spatial data pipelines, ML-driven features, database design, API architecture, and model integration.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript/TypeScript, C/C++

Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, async programming

AI / ML: TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, RAG, LlamaIndex, HuggingFace embeddings

Databases: PostgreSQL, MongoDB, MySQL, Firebase, Pinecone

Tools: Git, GitHub, Docker, Solidity, Hardhat, OpenCV, Matplotlib

EDUCATION

Air University Islamabad, Pakistan — Bachelor of Software Engineering

2022 – 2026

EXPERIENCE

AI / Machine Learning Intern

July 2025 – Present

WeatherWalayPK, Islamabad, Pakistan

- Built a high-performance FastAPI backend with asynchronous PostgreSQL connection pooling for 2.2M+ forecast records.
- Engineered IDW/Kriging spatial pipelines that convert scattered weather-station observations into structured geographic grids.
- Created compressed GeoTIFF disk caching with hot-swappable in-memory NumPy caches and automated hourly reprocessing when new forecast data arrives.
- Developed REST APIs for forecast retrieval, district filtering, model comparison, MAE, precision/recall, and JSON/GeoTIFF delivery.
- Fused PERSIANN, CHIRPS, and CMORPH satellite data with ground observations using resampling, bias correction, and quality control.
- Built 3D globe maps, district forecast grids, evaluation dashboards, and AI-generated alerts for the Weather X portal.

Frontend Developer Intern

June 2024 – August 2024

Rex Technologies, Lahore, Pakistan

- Built responsive React.js components, reusable UI flows, and REST API integrations for production web applications.

PROJECTS

FYP — SmartRent: Decentralized Rental Management Platform

TypeScript, React, Node.js, Solidity, MongoDB, Firebase, TensorFlow

RAG Chatbot — NLP

FastAPI, Pinecone, Groq LLM (LLaMA 3.1), HuggingFace embeddings, LlamaIndex

- Built a context-aware RAG chatbot with semantic retrieval in Pinecone, HuggingFace embeddings, and fast LLaMA 3.1 inference through Groq.

Spatial Weather Interpolation — Kriging & IDW

Python, Pandas, NumPy, Matplotlib, Jupyter Notebook

- Applied IDW and Kriging to interpolate missing Islamabad weather observations, then evaluated accuracy using MAE, residual analysis, and spatial visualizations.

IDW-Based 3D LUT Generator & Image Application

Python, OpenCV, NumPy

- Developed an interactive image-processing application that generates 3D lookup tables with IDW and applies real-time color transformations using OpenCV and NumPy.

CERTIFICATIONS

- [freeCodeCamp: JavaScript Algorithms and Data Structures \(2024\)](#)
- [Microsoft: Data Analysis and Visualization with Power BI \(2026\)](#)
- [IBM: Generative AI for Growth Marketing \(2026\)](#)
- [Google: AI Fundamentals \(2026\)](#)